

Universidade do Porto
Mestrado em Sistema de Informações Geográficas e Ordenamento do Território
Cartografia dos Riscos

Practical Work - Cartografia dos Riscos

Flood risk map in Aricanduva Watershed

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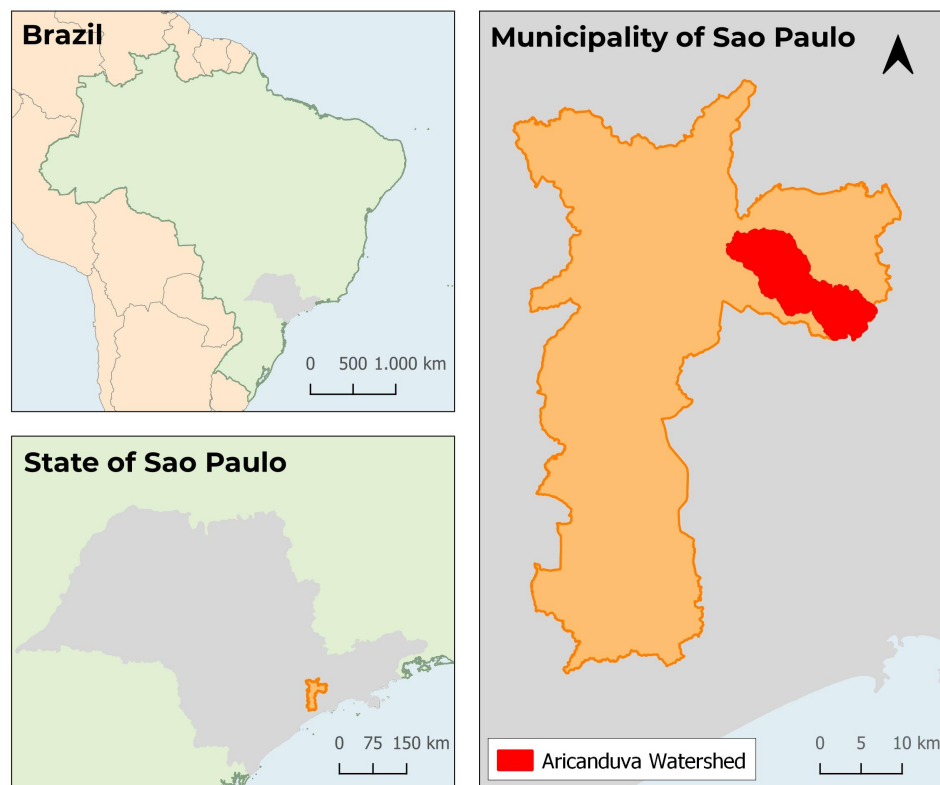
Introduction

Floods are a natural phenomena that have always posed dangers to lives, to the environment and to the economy. Even though they are naturally occurring, the process of rapid urbanization and the effects of climate change have led to an alarming increase in flood occurrences across many countries and have had a big impact on their severity and consequences. To address and mitigate these hazards, flood risk mapping plays a crucial role in understanding the potential risks associated with flood-prone urban areas. By accurately identifying vulnerable zones and assessing their susceptibility, decision-makers, urban planners, and emergency management agencies can develop effective strategies to reduce vulnerability and enhance community resilience.

Flood risk mapping integrates various data sources, including historical flood records, hydrological models, topographic information, and land use data. These inputs help generate flood hazard maps that illustrate areas at risk of inundation during different magnitudes of flood events. Analyzing these maps enables authorities to identify high-risk areas and prioritize them for flood mitigation measures, such as constructing flood control infrastructure, implementing zoning regulations, and developing early warning systems. Moreover, flood risk mapping informs land use planning and urban development practices.

Understanding the vulnerabilities of different areas allows for informed decision-making regarding the location and design of critical infrastructure. It also promotes the preservation of natural floodplain areas and encourages the implementation of sustainable drainage systems to mitigate the impacts of urban expansion on flood risk. This study aims to explore and analyze the flood risk within the Aricanduva basin.

Study Area



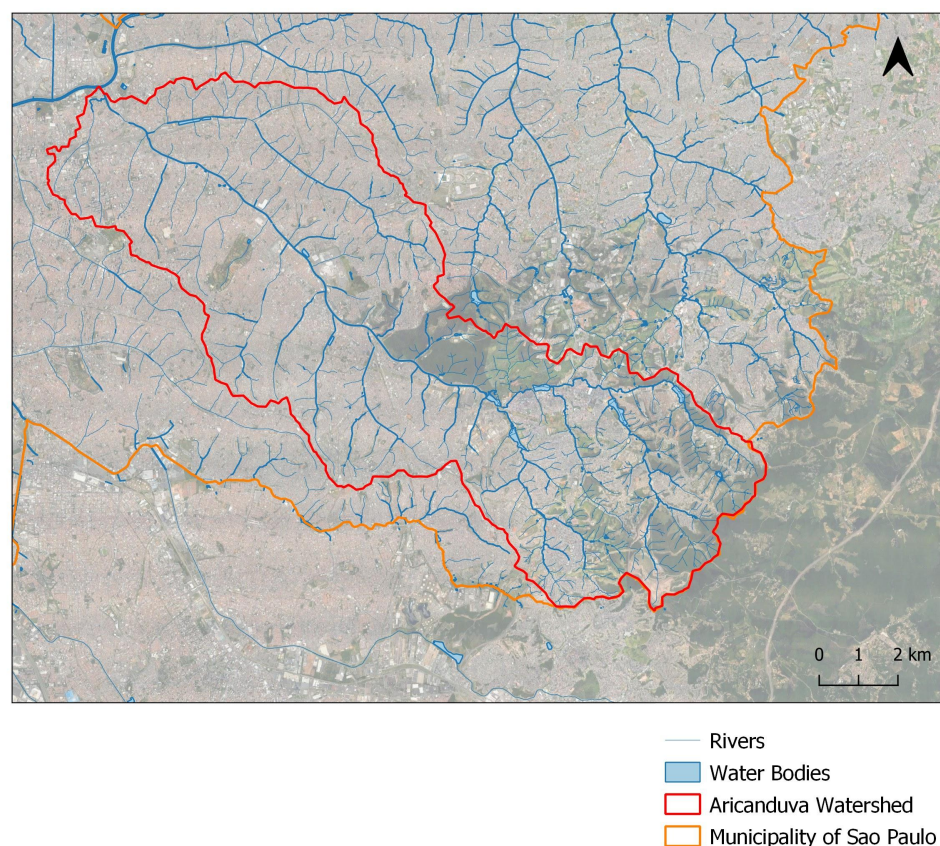
Sources: São Paulo limits - PMSP / Brazil - GADM / South America - Efrain Maps
Coordinate System: SIRGAS 2000 UTM Zone 23S

Figure 1. Location of the study area.

Aricanduva Basin is a sub-basin belonging to the Tietê River Basin. Tietê runs for 450 km and crosses the entire state of São Paulo, where the metropolitan region of São Paulo is situated. Even though the Tietê's headwaters are near the coast, in Serra do Mar - a mere 22 km from the ocean - and then runs in the opposite direction towards the hinterlands, being an affluent of Parana River, it encompasses a very important network of rivers on which the city of São Paulo was first established and has played a significant role in its development.

The São Paulo Metropolis, one of the largest urban agglomerations in the world, has experienced substantial growth over the years (URBANET, 2018). The process of urbanization that it has undergone is one due to industrialization and large settlements of migrant and immigrant populations in the 20th century, leading to rapid urban sprawl (Kogan, 2013). This resulted in the occupation of floodplains, either by the poorer sections of the population who couldn't afford living in less risk prone areas, or by the industry who took advantage of the large swathes of unoccupied land and energy provision, or as an easy path for road development, as is also the case of Aricanduva River. These unique characteristics of the Tietê river system make this basin susceptible to recurrent flood events, posing significant challenges for the surrounding communities and highlighting the importance of comprehensive flood risk management strategies.

The urban expansion has resulted in increased impervious surfaces which significantly alter the natural hydrological regime (McGrane, 2016). The transformation of land surfaces from permeable to impervious leads to rapid runoff during rainfall events, overwhelming the existing drainage systems and exacerbating the risk of flooding in low-lying areas.



Sources: Google Satellite / Sao Paulo limits - PMSP
Coordinate System: SIRGAS 2000 UTM Zone 23S

Figure 2. Urban fabric and Aricanduva River Basin.

The presence of open water channels within the Aricanduva basin also contributes to the flood risk. It is an area of historical recurrence of significant floods since its fast occupation in the 1970s (Simas, 2017), which was not accompanied by appropriate and well dimensioned infrastructure to mitigate its effects. These channels can become overwhelmed during intense rainfall and are significantly constricted by the urban fabric, leading to backflow and localized flooding. The effectiveness of these channels in managing floodwater depends on factors such as maintenance, sedimentation levels, and their integration into a comprehensive flood management system (Greco, 2014). Given the complex interplay of urbanization, hydrological characteristics, social vulnerability, and infrastructure considerations, flood risk mapping becomes an indispensable tool for understanding and addressing the flood risks in the Aricanduva basin.

Methodology

There are several possible methodologies utilized in flood risk mapping, varying in complexity and suitability for different available data and different scenarios. A brief research was conducted to compare our study to some of the methodologies utilized in our respective countries.

India

In the “*Impact of urban sprawl on future flooding in Chennai city, India.*”(2019) study, The catastrophic 2015 Chennai flood event is simulated and a similar extreme event is carried out for the predicted urban sprawl by 2030 and likely change in rainfall scenario.

The urban areas were flooded not only due to overflowing rivers but also due to lack of efficient drainage of the stormwater. It is thus difficult to separate fluvial and pluvial flooding extents and calibrate Manning’s n value. Hence, in this study, all the scenarios are run for the **Manning’s n values of 0.025, 0.03, 0.035 and 0.04m^{-1/3 s}** to take care of the uncertainty associated with the friction coefficient in the simulated results. The simulations are run with a stable and convergent time step of 6 s. In order to validate the baseline simulation, the flood-marks surveyed soon after the 2015 Chennai flood are used. The computed water depths are made to prepare flood inundation maps.

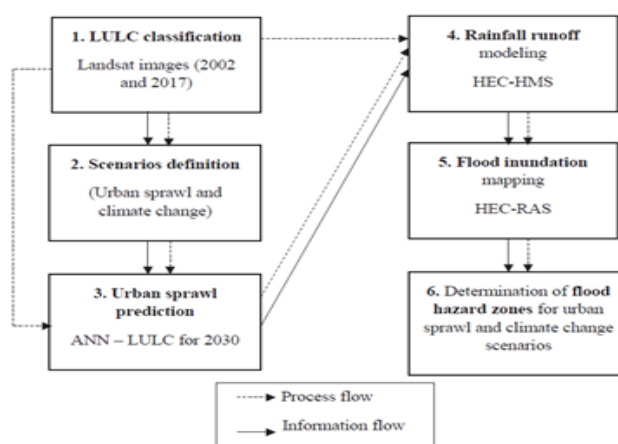


Figure 3. Framework involved in determination of flood hazard zones.

Extreme rainfall events on conveyance capacity of urban Storm Water Network (SWN) of Hyderabad City, India, along with flood risk analysis and inundation mapping were analyzed and the following methodology was formulated:

1. Nonlinear regression-based statistical downscaling to obtain future rainfall for Hyderabad City.
2. Employing Storm Water Management Model to quantify the performance of current SWN for historic and future extreme rainfall events.
3. Flood risk analysis and inundation mapping based on runoff resulting from historic and future extreme rainfall events.

For the purpose of this analysis, the probability of flooding was dependent only on rainfall. Annual flood risk (R) is determined based on the probability of flood to occur (P) and the consequences of flooding (C).

$$R = C \times P$$

Consequences of flooding (C) were calculated for each storm water zone. C is a product of value of assets (V), susceptibility to damage (S) and exposure to flooding (E).

$$C = V \times S \times E$$

Malaysia

A case study of Sg. Kelantan river basin in Malaysia (Ghazali and Osman, 2019) was referred to. The Sg. Kelantan river basin in Malaysia shares topological similarities with the study area and experiences flooding due to the narrowness of the river at its lower reaches. While both studies collected data on topography, hydrological, river morphology, spatial and land use, the study in Sg. Kelantan included future land use scenarios. The Sg. Kelantan study plotted double mass curves while this study used Thiessen Polygons to identify the relevant precipitation stations.

The hydrological modeling in Sg. Kelantan utilized InfoWorks Integrated Catchments Model (ICM) Software, with the SCS Synthetic Unit Hydrograph Method to calibrate extreme rainfall events and flood frequency analysis. The design rainfalls obtained were then applied to the calibrated rainfall-runoff model to produce flood hydrographs of various return periods, and were later used as input for boundary conditions of the hydraulic model. In our study, Inverse Distance Weighting (IDW) approach was used to complete the missing data. The yearly maximum precipitation from 1973 to 2003 was applied in Gumbel distribution for a return period of 100 years. The time of concentration (TC) and runoff peak discharge (Qp) generated from Giandotti's formula served as input for boundary conditions of the hydraulic model.

In terms of hydraulic modeling, InfoWorks ICM and IBER were used in the Sg. Kelantan and this study, respectively. Both studies incorporated bridges and other features along the river to simulate the real conditions. In our study, the cells of the bridge areas were set to no data and the Elevation Void Fill Function in Arcmap, which employs a combination of plane fitting and Inverse Distance Weighting (IDW) was utilized, to accurately represent the flow conditions under the bridges. Overall, accurate data collection and modeling techniques are vital as they determine the accuracy and reliability of the flood inundation simulation maps.

Brazil

For the Aricanduva watershed we referred to "Caderno da Bacia Hidrográfica do Aricanduva" (PMSP, 2022) and "Análise retrospectiva de episódios de inundações na Bacia Hidrográfica do Rio Aricanduva - São Paulo" (Simas, 2017). The first is the official management report from the city's watershed committee and the second is an academic thesis focusing on the retrospective analysis of two historical periods of significant floods in the basin.

Análise retrospectiva de episódios de inundações na Bacia Hidrográfica do Rio Aricanduva - São Paulo, 2017

In this study, the author utilized rainfall information from pluviometric stations relevant to the area, also making use of Thiessen Polygons to determine their contribution to the discharge. Because this work aimed to look retrospectively at some seasons, it considered different aspects of the data, instead of the entire series, as well as the entire basin: total precipitation for the rain season (November to March), number of rainy days, highest daily precipitation, average rainy day precipitation and highest total accumulated precipitation in an uninterrupted series. Then, he used the Quantis technique to classify the information and obtain the seasons to be studied. The missing data, when existed, was interpolated using kriging. The spatial model was then created taking into account other variables such as slope, litology, land use, hydrodynamic trends and social and spatial vulnerability, which was then compared to a reconstruction of the original morphological terrain and the documented effects of the events.

Caderno da Bacia Hidrográfica do Aricanduva, 2022

Considering that the area is a highly urbanized part of a Metropolitan Region and has already undergone several interventions to mitigate the risks and effects of floods, this work is far more complex and comprehensive than what we have attempted, aimed at looking further on how to improve flood management. First, as with the present study, the precipitation was analyzed based on meteorological radar data from the telemetric network. However, they considered not only the return period (for 2, 5, 10, 25 and 100 years), but also the critical event duration. They utilized the formulae:

$$i_{t_d, Tr} = A(t_d + B)^C + D(t_d + E)^F \left\{ G + H \ln \left[\ln \left(\frac{Tr}{Tr - 1} \right) \right] \right\}$$

valid for $10 \leq t \leq 1440$ min, where:

A = 32,77

B = 20

C = - 0,878

D = 16,1

E = 30

F = - 0,9306

G = - 0,4692

H = - 0,8474

td is the rain duration, in minutes;

Tr is the return period, in years;

It_{d,Tr} is the rain intensity, in mm/min, for the duration td (min) and the return period Tr (years)

The resulting total precipitation accumulated in 6 hours of uninterrupted rainfall for a return period of 100 years was 141,2 mm, which is very similar to our finding of 140,85 mm of maximum daily precipitation for a return period of 100 years, although, in our study, this indicated a time of concentration of 10.26 hours.

The river basin was then divided into 47 sub-basins to calculate the runoff considering the tributary rivers contributions, the drainage network, the land use legislation and the current percentage of impervious area of each sub-basin. The discharge was estimated using the SWMM (Storm Water Management Model), developed by the Environmental Protection Agency, in the PCSWMM graphical interface in Windows environment. For the infiltration, they utilized the CN methodology, developed by the Soil Conservation Service. The model uses the Dynamic Wave method, which solves the complete Saint-Venant's equations for the study of surface runoff. Finally, the flood information was combined with population density and vulnerability indexes to create the risk analysis using the formulae:

$$R = H \times P \times V$$

Then, we did the following hydrology process to determine the watershed:

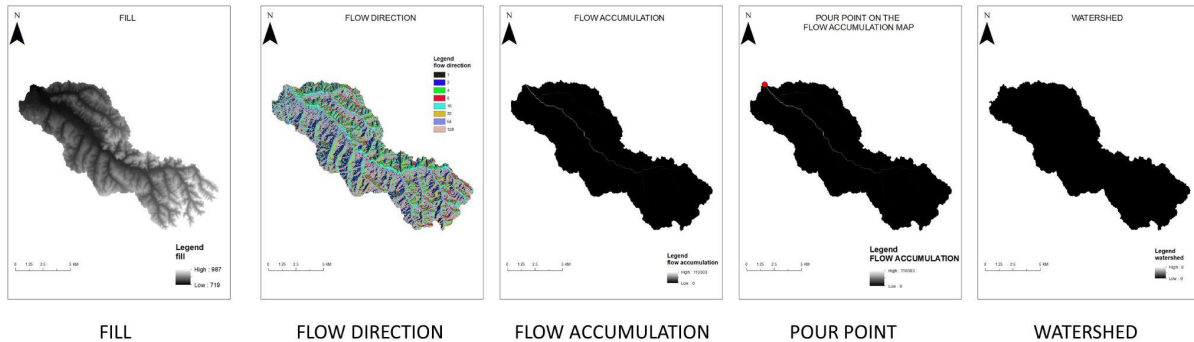


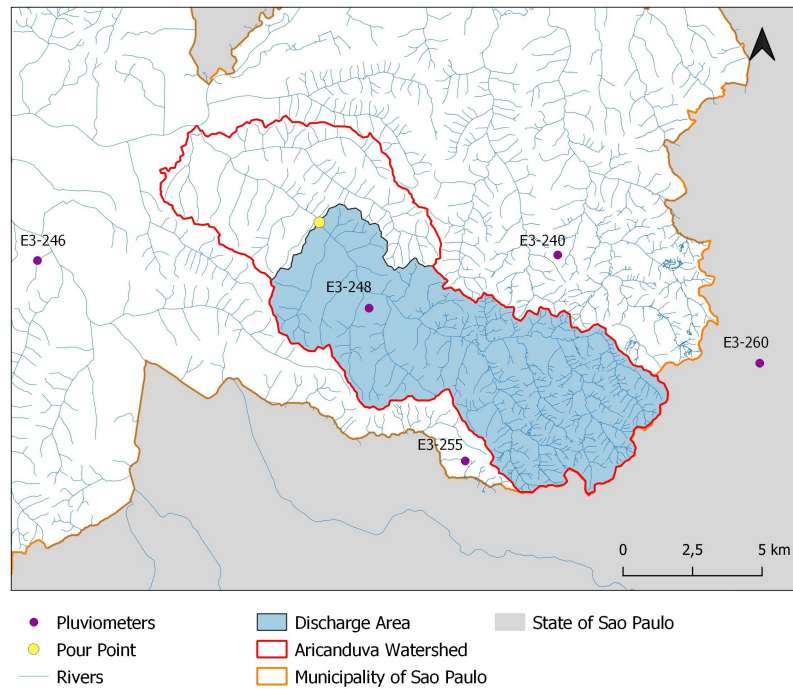
Figure 6. Hydrology process done in ArcMap.

- **Fill:** Fills sink in a surface raster to remove small imperfections in the data. Fill can also be used to remove peaks. To create a Fill, A surface raster is required which is the DEM.
- **Flow Direction:** Creates a raster of flow direction from each cell to its downslope neighbor, or neighbors.
- **Flow Accumulation:** Creates a raster of accumulated flow into each cell. The result of Flow Accumulation is a raster of accumulated flow to each cell, as determined by accumulating the weight for all cells that flow into each downslope cell. Output cells with a high flow accumulation are areas of concentrated flow and can be used to identify stream channels. To create the flow accumulation, the flow direction raster is taken as the input file keeping the flow direction as D8 type.
- **Pour Point:** It is the point on the surface where the water flows out of an area. It is used for determining the Watershed of an area. A point shapefile was created.
- **Watershed:** The value of each watershed will be taken from the value of the source in the input raster or feature pour point data. For the creation of Watershed, we require the flow direction raster along with the pour point data as mentioned above.

Data Collection - Precipitation

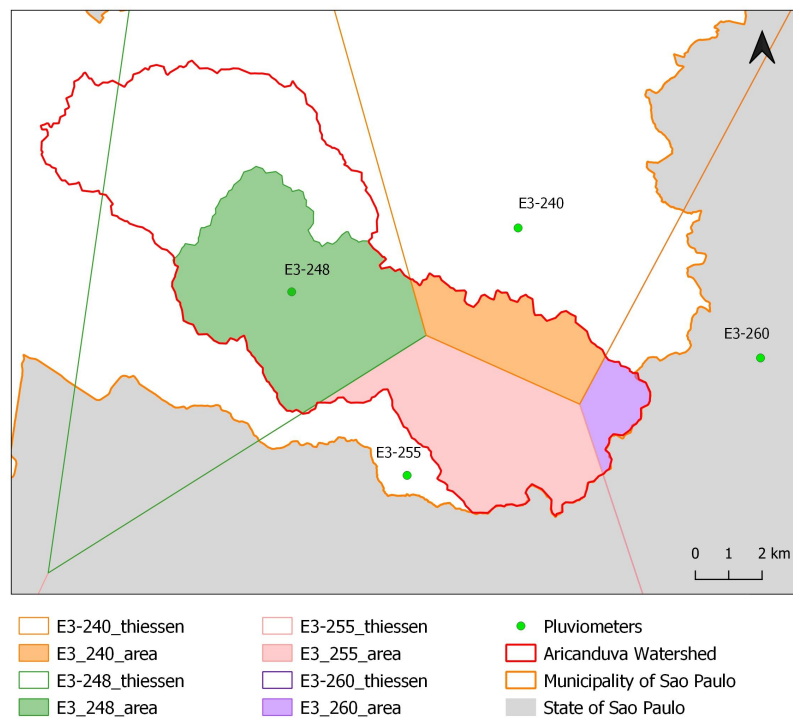
Once we had the confirmed limits of the watershed, we needed to determine the portion of it that contributes to the area we want to analyze: the discharge area. It is the area upstream from the pourpoint, located 5 km upstream to where Aricanduva meets Tietê River. We also needed to select which pluviometers were relevant to our area. We needed to define the time of concentration (TC) and runoff peak discharge (Qp) for a return period of 100 years at the pour point defined in Figure 7. These values were used as input parameters for the boundary conditions in our flood simulation. We used Giandotti's formula to calculate these values.

In order to achieve this, we collected 30 years (1973 to 2003) of precipitation data from the pluviometers relevant to the study area. However, there are information gaps in the precipitation data obtained from pluviometers E3-240, E3-248, E3-255, and E3-260 during this time period, as these stations were not operational for the entirety of this period. E3-240 operated from 1972 to 1994, E3-248 operated from 1973 to 1996, E3-255 operated from 1979 to 2003, and E3-260 operated from 1984 to 2015.



Sources: Pluviometers - DAEE / Sao Paulo limits - PMSP / Rivers - FCHT and SMUL
 Coordinate System: SIRGAS 2000 UTM Zone 23S

Figure 7. Discharge area, pour point and pluviometers.



Sources: Pluviometers - DAEE / Sao Paulo limits - PMSP
 Coordinate System: SIRGAS 2000 UTM Zone 23S

Figure 8. The stations' Thiessen Polygons signaling their contribution area to the discharge area.

To overcome this limitation, we applied Inverse Distance Weighting (IDW) to interpolate the missing data using the precipitation measurements from nearby stations. Spatial interpolation methodologies like Kriging and Multiple Regression were not applicable due to the limited data available, as they required more variables from the stations. Additionally, there is only one station in each Thiessen Polygon while the missing data would be estimated based on the average of the neighboring stations that fall within the same polygon in this methodology. Hence, IDW is more applicable to our scenario. IDW is a spatial interpolation methodology that involves assigning weights to the neighboring stations based on their distance from the station with the missing data. The closer the station, the higher the weight. The missing data is then estimated based on the weighted average of the neighboring stations.

OBJEC TID	INPUT_ FIELD	NEAR_ FID	DISTANCE(KM)	WEIG HT
1	E3-248	E3-255	6.50	0.0237
2	E3-248	E3-260	14.19	0.0050
3	E3-248	E3-246	12.05	0.0069
4	E3-248	E3-240	7.05	0.0201
5	E3-240	E3-255	8.13	0.0151
6	E3-240	E3-260	8.25	0.0147
7	E3-240	E3-248	7.05	0.0201
8	E3-240	E3-246	18.71	0.0029
9	E3-255	E3-260	11.16	0.0080
10	E3-255	E3-248	6.50	0.0237
11	E3-255	E3-246	17.00	0.0035
12	E3-255	E3-240	8.13	0.0151
13	E3-246	E3-255	17.00	0.0035
14	E3-246	E3-260	26.24	0.0015
15	E3-246	E3-248	12.05	0.0069
16	E3-246	E3-240	18.71	0.0029
17	E3-260	E3-255	11.16	0.0080
18	E3-260	E3-248	14.19	0.0050
19	E3-260	E3-246	26.24	0.0015
20	E3-260	E3-240	8.25	0.0147

Table 1. Distance between each station.

The distance between the stations was calculated using Point Distance tools in ArcMap. Pluviometer E3-246 was taken into account for the purposes of filling the data gaps due to data limitation but not for the calculation of the maximum daily precipitation, as the station fell outside the Thiessen Polygon of our study area. After determining the distance between the stations, we assigned weights to each neighboring station based on their distance from each station, with closer stations having a higher weight.

$$\text{Station } n \text{ weight} = 1/D^p$$

D = distance from the station of missing data

p = power parameter that determines the rate at which the weights decrease with distance

We then calculated the weighted average by multiplying the precipitation value for each neighboring station by its weight, and dividing the sum of these values by the sum of the weights, to estimate the missing precipitation value. After the missing precipitation data was filled, we defined the maximum precipitation of each year and identified the average and standard deviation of the maximum precipitation from 1973 to 2003 for input into the Gumbel distribution.

Precipitation/Pluviometer	E3-240	E3-248	E3-255	E3-260
Max	115.90	190.10	134.80	145.10
Min	39.80	37.20	39.00	45.30
Average	70.05	79.51	75.00	79.30
Std Deviation	17.50	27.51	23.45	24.14

Table 2. The precipitation values from 1973 to 2003.

Discharge Calculations

Gumbel's (2004) probabilistic model

In probability theory and statistics, the Gumbel distribution (also known as the type-I generalized extreme value distribution) is commonly used to model the distribution of extreme values, particularly the maximum or minimum of several samples of various distributions. Gumbel distribution is widely used in hydrological and meteorological studies for the prediction of flood peaks, maximum rainfall etc.

There are different variations and modifications in the formulas and parameterizations used by researchers and applications, depending on their specific research purposes and requirements. In this study, we chose the Gumbel model based on its suitability for our available data and research objectives. We derived the average and standard deviation values of the maximum daily precipitation from pluviometer E3_240, E3_248, E3_255, and E3_260, to calculate the maximum daily precipitation value for a return period of 100 years, which will be used as an input in Giandotti formula for estimating the time of concentration (TC) and runoff peak discharge (Qp).

$$X = \bar{X} - S \left\{ 0,45 + 0,7797 \ln \left[\ln \frac{Tr}{Tr - 1} \right] \right\}$$

- $\bar{X}$ – average of the series of maximum daily precipitation values in the year
- S – standard deviation of the series of maximum daily precipitation values in the year
- TR – return time

Gumbel Distribution		
E3_248	Jardim Centenario	124.96
E3_240	Itaquera	165.82
E3_255	Parque São Rafael	148.60
E3_260	Guaio	155.00

Table 3. The maximum daily precipitation value for a return period of 100 years for each station

Giandotti – Runoff Peak Discharge (Qp)

$$Q_p = \lambda * \text{Area} * h / TC$$

- λ – constant that will vary depending on the area of the watershed
- Area – area of the watershed
- h^* – the maximum daily precipitation resulted from the Gumbel's formula (for the chosen return period)
- $^*h = (\Sigma (\text{proportion of contribution area of station} * \text{Gumble distribution of station})) / 100$
- TC – time of concentration

Giandotti (1934) – Time of Concentration (TC)

$$TC = (\sqrt[4]{A} + 1.5L) / 0.8\sqrt{H}$$

- A – area of the watershed (km²)
- L – length of the entire watershed (km)
- H – the altmetric amplitude (m)

Giandotti - Runoff Peak Discharge

$$325\text{m}^3/\text{sec} = 0.345 * 68.62^A * 140.85^B / 10.26^C$$

Note A:

Proportion of Contribution Area		Area Km2	%
Discharge area	Medio e alto Aricanduva	68,62	1
E3_248	Jardim Centenario	31,41	0,46
E3_240	Itaquera	10,85	0,16
E3_255	Parque São Rafael	22,6	0,33
E3_260	Guaio	3,76	0,05
	Total	68,62	1,00

Note B:

Max daily precipitation for a return period of 100 years

$$140.85 \text{ mm} = ((45.77 * 124.96) + (15.81 * 165.82) + (32.94 * 148.6) + (5.48 * 155)) / 100$$

Note C:

Giandotti - Time of concentration

$$10.26 \text{ hours} = (\sqrt[4]{68.62^A} + 1.5(24.97)) / 0.8\sqrt{74}$$

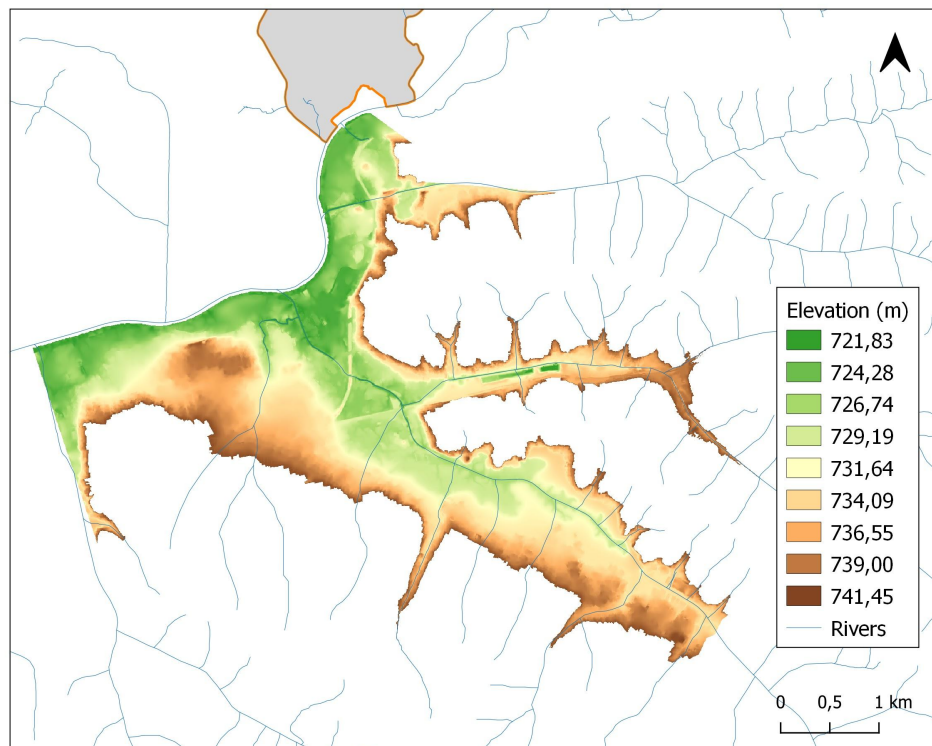
Table 4. Calculation for Giandotti - Runoff Peak Discharge, Proportion of Contribution Area, Max daily precipitation for a return period of 100 years, and Giandotti - Time of concentration

The maximum daily precipitation (h) result from the Gumbel's formula for 100 years is 140.85 mm^B. Based on the calculation, as in Table 4, we derived the TC value of 10.26 hours and Qp value of 325 m³/sec, to use as input parameters for the boundary conditions in our flood simulation.

IBER Model

IBER is a powerful two-dimensional numerical model for hydraulic and hydrological modeling that can be used to simulate and analyze river flows and flooding events. It is developed by a collaboration of Universidade da Coruña, Universitat Politècnica de Catalunya, and CEDEX. In this study, we used IBER to simulate the river flood inundation in our study area, mapping the areas that are likely to be affected during the runoff peak discharge.

After a first simulation, we realized that it could be necessary to include the floodplains around Tietê River that are nearby but not part of the Aricanduva Basin, as they could be affected. For that, we created a new Digital Elevation Model with 1m resolution.



Sources: Contour - PMSP
Coordinate System: SIRGAS 2000 UTM Zone 23S

Figure 9. Hypsometry for the simulation area.

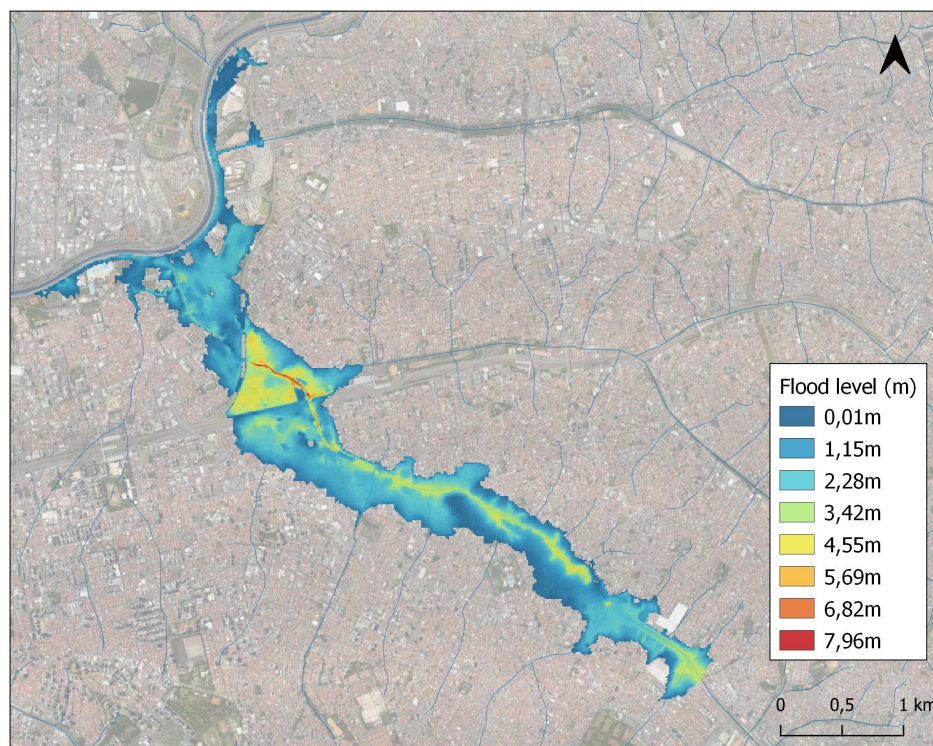
For the new simulation, first, we converted the Digital Elevation Model (DEM) using the Raster to ASCII tool in Arcmap, and then created RTIN in IBER. We imported DEM.txt as mesh to IBER, and set the tolerance at 0.25 m to define the maximum acceptable distance between the output triangulation and the input points, maximum side at 100 m to prevent overly large triangles from being created, which can lead to inaccurate results, and minimum side at 5 m to avoid excessively detailed RTIN files.

For boundary conditions of hydrodynamics, we assigned the parameters for 2D inlet and outlet. For the inlet, we set total discharge, critical/subcritical for condition, and discharge value of 325 m³/sec (from Giandotti's formula, Table 4). Additionally, we assigned it to the pour point and drew a 2D inlet. For the 2D outlet, we set the flow condition to supercritical/critical, assigned it to the downstream, and drew the entire mesh for all 2D analysis.

For the roughness, we selected "continuous urban fabric" reflecting the most prevalent characteristic of our study area. We assigned the elements to the entire area, and drew it for all materials. In the problem data, we created a new simulation with maximum simulation time of 36,000 s to reflect the 10.26 hours of TC we derived from Giandotti's formula (refer to Table 4), and the results 2D time interval of 3,600 s, as well as setting Hazard RD9/2008 and ACA for the results. Finally, we generated the river flood inundation simulation for our study area, as shown in Figure 10.

Results and Discussion

The resulting flooded area is quite significant, particularly when considering that a large portion of the area is occupied by human activities. As we know, this urbanization is the result of an exponential urban growth that disregarded the actual natural floodplains and has seen its effects on at least two significant historical floods.



Sources: Google Satellite
Coordinate System: SIRGAS 2000 UTM Zone 23S

Figure 10. Simulated flooded area for a return period of 100 years.

The maximum daily precipitation of 140.85 mm for a return period of 100 years signifies the highest amount of rainfall expected to occur within a single day in an extreme weather event within a 100-year time span. Because of the difficulty in accessing more contemporary data from pluviometers currently in use, the study was based on a series from 1973 to 2003, which means that probable

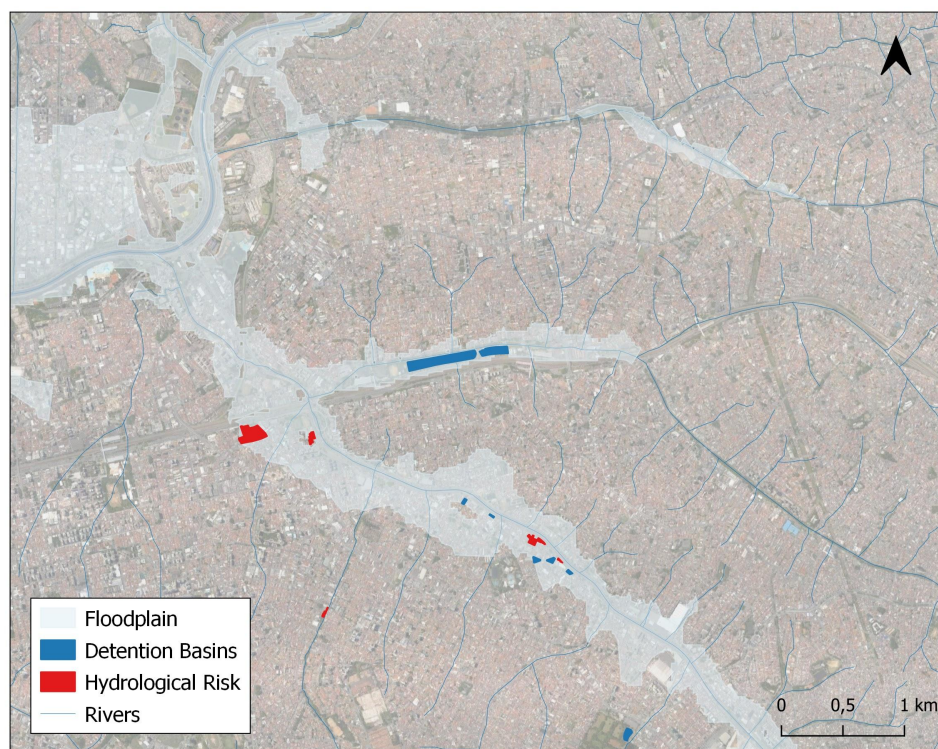
recent variations in the precipitation patterns were not contemplated, meaning there is a possibility the value could be higher for future events.

The time of concentration of 10.26 hours represents the duration it takes for water to travel from the most distant part of the watershed to the outlet. In the case of the Aricanduva basin, this relatively long period is explained by the relatively low elevations in most of the basin, as seen in Figure 5, and the large floodplains are expected, being that this low-level final section has historically been an area of concentration.

However, the impervious nature of most of the studied surface compromises drainage, and according to Simas (2017), the areas with higher propensity for infiltration have been reduced in more than 99% (in comparison to the original environment) and the areas with more propensity to generate runoffs have increased in more than 18000%. Because our study doesn't take some elements into account, such as the existing drainage system or the basin's lithology, we can see that a large problem area is indicated in the triangular site created by the embankments for the road and rail system, even though this area is not particularly highlighted in the current drainage management plans, as we'll see shortly.

It is crucial to also acknowledge the limitations of our study when it comes to the limited access to data on urban drainage and sewage systems, as well as specific infrastructure measures, such as tunnels, retention basins, polderes etc, which were not included in the study and would alter the complexity of the entire simulation. These elements can impact flood patterns by redirecting or containing water, even though they may not entirely replace the natural floodplain function.

To address these limitations and validate our findings, we compared our results with the findings from official studies.



Sources: Google Satellite / Areas susceptible to floods - SIURB and FCTH / Hydrological Risk - SMSU / Detention Basins - FCTH and SMUL
Coordinate System: SIRGAS 2000 UTM Zone 23S

Figure 11. Data compiled from Cadernos de Bacia Hidrográfica (SIURB/FCTH) and from Plano Diretor de Macro drenagem - PDMAT3 (DAEE), as of 2020.

It is interesting to see the results the City of São Paulo has achieved in its preliminary studies for its drainage master plan and the earlier versions of the Cadernos de Bacia Hidrográfica, which can be seen in Figure 11. The flood prone areas are very similar, which means that, as an initial diagnosis, the methodology we used can bring relevant results. The same problem area identified in our study is replicated here and it is expanded towards the existing detention basin, which didn't occur in our simulation. Unfortunately, we haven't been able to locate the methodology and calculations used, to better compare them.

We also compared our results to the ones from the “Caderno de Bacia Hidrográfica: Bacia do Córrego Aricanduva” (PMSP, 2022), as seen in Figure 12, which was briefly reviewed in this study's methodology. In this case, the flooded areas are significantly smaller and concentrate in different points. These differences are expected, considering this study computes more variables and aims to further equip the area to mitigate flood risks.

FIGURA 4.1 Mapeamento das áreas inundáveis na bacia do Rio Aricanduva

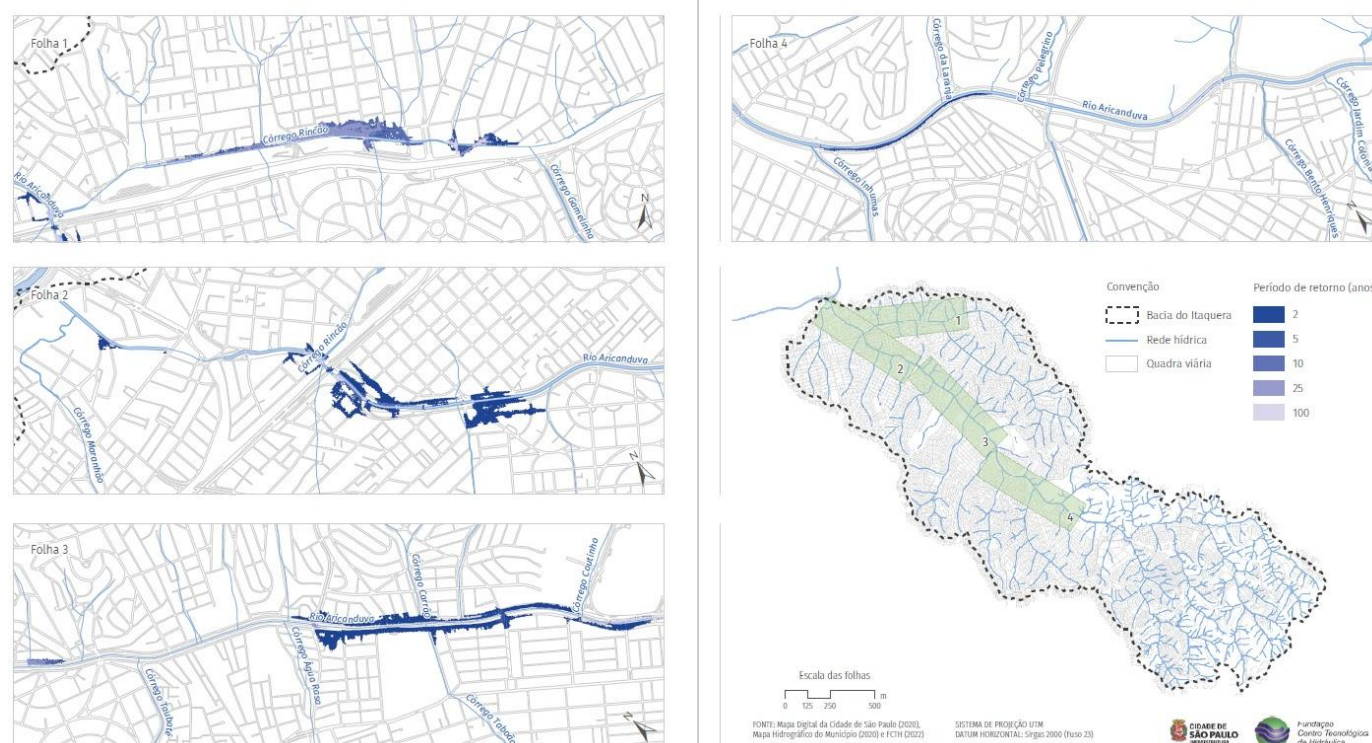


Figure 12. Simulated flooded areas for different return periods, Caderno de Bacia Hidrográfica: Bacia do Córrego Aricanduva (PMSP, 2022).

Conclusions

There is considerable difference between what would be that semi-natural flow of the water without any flood preventive measures, human occupation and reality. A methodology such as the one used in our study can be sufficient for areas that have less human-made alterations, providing a comprehensive understanding of the rivers's flow patterns, to determine its floodplains and mitigating measures. With more detailed information on land use and a more detailed Digital Elevation Model, we could have reached an even better result in this regard, but it would still be a first step.

In such an urbanized area as the one studied, it is important to consider the limitations and applicability of the simulation. In our study area, the magnitude of the urban fabric in and around the

natural floodplain alters the flow patterns, the soil capacity for water infiltration, the speed of the runoff, among other parameters, which exacerbates flood risks and its effects on human lives and livelihood.

In a context where these issues have been semi-addressed by drainage infrastructure, although incomplete and insufficient, maintaining such infrastructure, reexamining conditions and proposing solutions that go beyond the tried and tested is a must for designing effective flood prevention and mitigation strategies. It is unrealistic to assume we could completely revert the conditions to a more natural and acceptable scenario, and it is equally unrealistic to expect that conventional engineering solutions such as detention basins and underground channels could be enough. In a metropolis like São Paulo, one has to work with the conditions that are given, for any proposal has to be considered within a very complex existing context.

References

- Albrigo, NS; Pereira, MTS; Soares, NTD; Andrade, GS; Souza, VAS; Rosa, ALD (2021) *Determination of the intensity-duration-frequency curve in Cruzeiro do Sul – Acre*. Science and Nature, Santa Maria, v. 43, e30, p. 1–13, 2021. Retrieved from: <https://doi.org/10.5902/2179460X38301>, 19.5.2023.
- Ghazali, IHNHM; Osman, S. (2019) *Flood Hazard Mapping in Malaysia: Case Study Sg. Kelantan river basin*. Department of Irrigation and Drainage, Malaysia. Retrieved from: http://hywr.kuciv.kyoto-u.ac.jp/ihp/cha/vol1/Malaysia_HazardMap.pdf, 30.2.2023.
- Greco, S. E. (2014) *Ecological design of multifunctional open channels for flood control and conservation planning*. Landscape and Urban Planning Volume 131, November 2014, Pages 14-26. Retrieved from: <https://doi.org/10.1016/j.landurbplan.2014.07.002>
- Gómez, LC; Castellet, EB; Ramos, MS; Pita, MB; Alonso, AM (2019) *Iber applications basic guide*. ISBN: 978-84-9749-717-6. Retrieved from: <https://doi.org/10.17979/spudc.9788497497176>
- Kogan, G. (2013) *The Socio-Environmental History of The Floods In São Paulo 1887-1930*. UNESCOIHE . May 2013 MSc Thesis WM WSM 13.18. Retrieved from: https://cosmopista.files.wordpress.com/2014/11/thesis_gabrielkogan.pdf
- McGrane, S. J. (2016) *Impacts of urbanization on hydrological and water quality dynamics, and urban water management: a review*. Hydrological Sciences Journal Volume 61, 2016 - Issue 13. Retrieved from: <https://doi.org/10.1080/02626667.2015.1128084>
- N. Nithila Devi, B. Sridharan, Soumendra Nath Kuiry (2019), *Impact of urban sprawl on future flooding in Chennai city, India* Journal of Hydrology Volume 574, July 2019, Pages 486-496
- Prefeitura do Município de São Paulo (2022) *Caderno de bacia hidrográfica: bacia do córrego Aricanduva*. Fundação Centro Tecnológico de Hidráulica. 2a edição – São Paulo: FCTH/SIURB, 2022. Retrieved from: <https://www.prefeitura.sp.gov.br/cidade/secretarias/obras/noticias/?p=338531>
- Simas, Y. T. H. (2017) *Análise retrospectiva de episódios de inundações na Bacia Hidrográfica do Rio Aricanduva - São Paulo*. Master Thesis, FFLCH -USP. São Paulo, 2017. Retrieved from: <https://teses.usp.br/teses/disponiveis/8/8135/tde-21072017-162915/pt-br.php>
- Swathi Vemula, K. Srinivasa Raju, S. Sai Veena, A. Santosh Kumar (2018), *Urban floods in Hyderabad, India, under present and future rainfall scenarios: a case study*. Published online: 24 October 2018 Springer Nature B.V. 2018, Natural Hazards (2019) 95:637–655
- URBANET (2018) *Infographics: Urbanisation and Urban Development in Brazil*. Retrieved from: <https://www.urbanet.info/urban-development-brazil/>, 16.5.2023.